

花老师班补充作业 04.

$$13. \sum_{n=1}^{\infty} \frac{2n+1}{n!} x^n = 2 \sum_{n=1}^{\infty} \frac{1}{(n-1)!} x^n + \sum_{n=1}^{\infty} \frac{x^n}{n!} = 2x \sum_{n=1}^{\infty} \frac{1}{(n-1)!} x^{n-1} + \sum_{n=1}^{\infty} \frac{x^n}{n!}$$

$$= 2xe^x + e^x - 1$$

$$\text{let } x=1 \Rightarrow \sum_{n=1}^{\infty} \frac{2n+1}{n!} = \underline{\underline{3e-1}}$$

$$17. f(x) = \frac{x}{2+x-x^2} = \frac{x}{-(x+1)(x-2)} = \frac{\frac{2}{3}}{2-x} - \frac{\frac{1}{3}}{1+x}$$

$$= \frac{2}{3} \cdot \frac{1}{1-(x-1)} - \frac{\frac{1}{3}}{1+(x-1)} = \frac{2}{3} \cdot \frac{1}{1-(x-1)} - \frac{1}{6} \cdot \frac{1}{1+\frac{x-1}{2}}$$

$$= \sum_{n=0}^{\infty} \left[\frac{2}{3} - \frac{1}{6}(-1)^n \cdot \frac{1}{2^n} \right] (x-1)^n, \quad |x-1| < 1 \Rightarrow 0 < x < 2$$

$$40. \int_0^x \frac{\sin t}{t} dt = \int_0^x \frac{1}{t} \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!} dt = \int_0^x \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n+1)!} dt$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)(2n+1)!}$$

$$41. f(x) = \frac{1}{x^2-5x+6} = \frac{1}{(x-2)(x-3)} = \frac{1}{x-3} - \frac{1}{x-2} = \frac{1}{1+(x-4)} + \frac{1}{4+(x-4)}$$

$$= \frac{1}{1+(x-4)} + \frac{1}{2} \cdot \frac{1}{1+\frac{x-4}{2}} = \sum_{n=0}^{\infty} (-1)^n \left[1 + \frac{1}{2} \cdot \frac{1}{2^n} \right] (x-4)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(1 + \frac{1}{2^{n+1}} \right) (x-4)^n, \quad \begin{cases} |x-4| < 1 \\ \Rightarrow 3 < x < 5 \end{cases}$$

$$45. (1) \lim_{x \rightarrow 0} \frac{\sin x \ln(1+x) - x^2}{x^3} = \lim_{x \rightarrow 0} \frac{(x - \frac{x^3}{3!} + \dots)(x - \frac{x^2}{2} + \dots) - x^2}{x^3} = \frac{1}{2}$$

$$60. (1) \lim_{x \rightarrow 0} \frac{\sin ax - \sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{(ax - \frac{a^3 x^3}{3!} + \dots) - (x - \frac{x^3}{3!} + \dots) - x}{x^3} \Rightarrow \begin{cases} a-2=0 \\ \frac{a^3}{6} + \frac{1}{6} = b \end{cases} \Rightarrow \begin{cases} a=2 \\ b=-\frac{7}{6} \end{cases}$$